M3 DEMO · INTRODUCTION TO SOFTWARE ENGINEERING (15031001)

MindPay

Know how many days your money will last.

Team 13. Socrates and Skeletons

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PROBLEM

Logging is tedious, so records stay incomplete and money runs out before the month ends

Who

Students and first jobbers who pay by mobile banking many times a week

Why they skip it

Retyping every detail and hunting for payment slips

What it costs

They see the balance, not how fast it is going, and react too late

Our idea: log with almost no typing, then warn before the balance gets too low.

SCOPE

Six requirements, all built: what each one does

FR-1 Transaction management

Add, edit, categorize and delete income or expenses. Delete asks first and can be undone. Each user sees only their own records.

FR-2 Overview dashboard

Home shows balance, income, expenses and spending by category for today, 7 days, a month, 6 months or a year. Only confirmed records count.

FR-3 Voice logging

Say "rice 50 baht, water 15". The app turns speech into items you can edit, understands Thai number words and "yesterday", and saves only what you confirm.

FR-4 Automatic slip detection

Finds payment slips in the gallery and reads amount, date, time and bank. Readings at 80% confidence or more are counted; the rest wait as drafts. Duplicates are blocked.

FR-5 AI coach "Nong Kla"

Explains spending in plain Thai, as text and speech, using confirmed numbers only. Answers questions like "Can I buy this?" and falls back to a rule-based summary if AI is down.

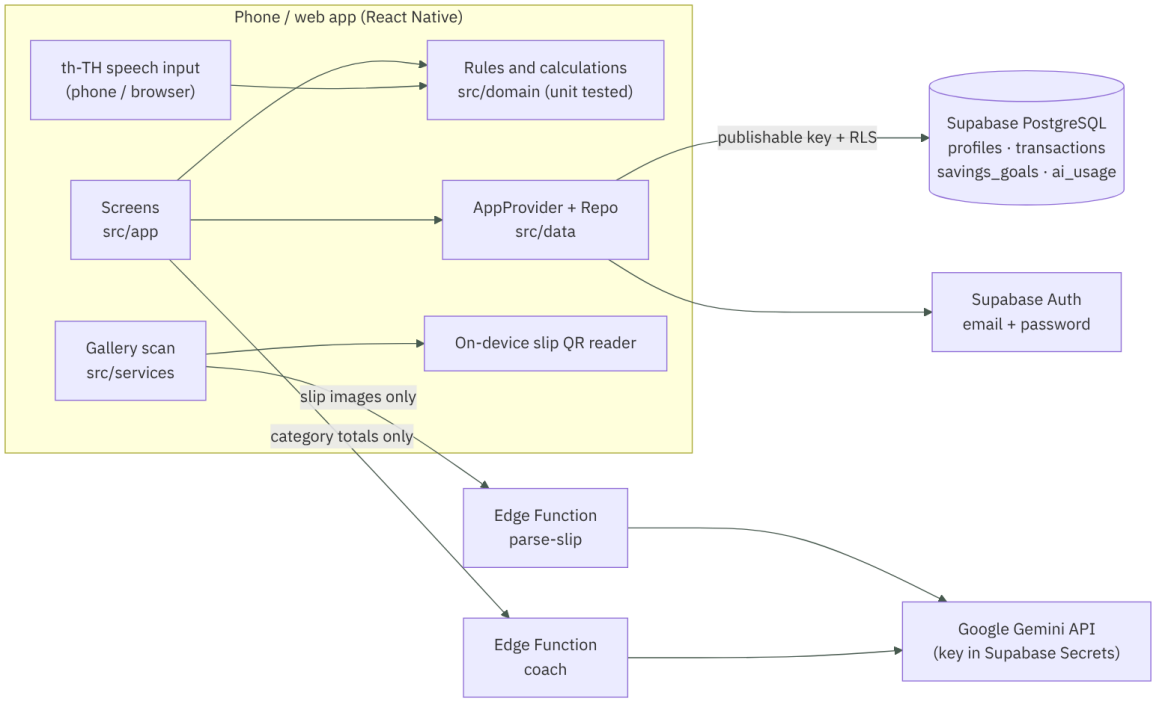
FR-6 Money Runway

Estimates the day the balance reaches the user's low-balance line from the last 7 days of spending, and shows how cutting spending adds days.

FR-7 Accountability Mode and FR-8 Online mode from the original M1 Charter and SRS are planned for a later release.

DESIGN

Architecture



One React Native app for Android and web

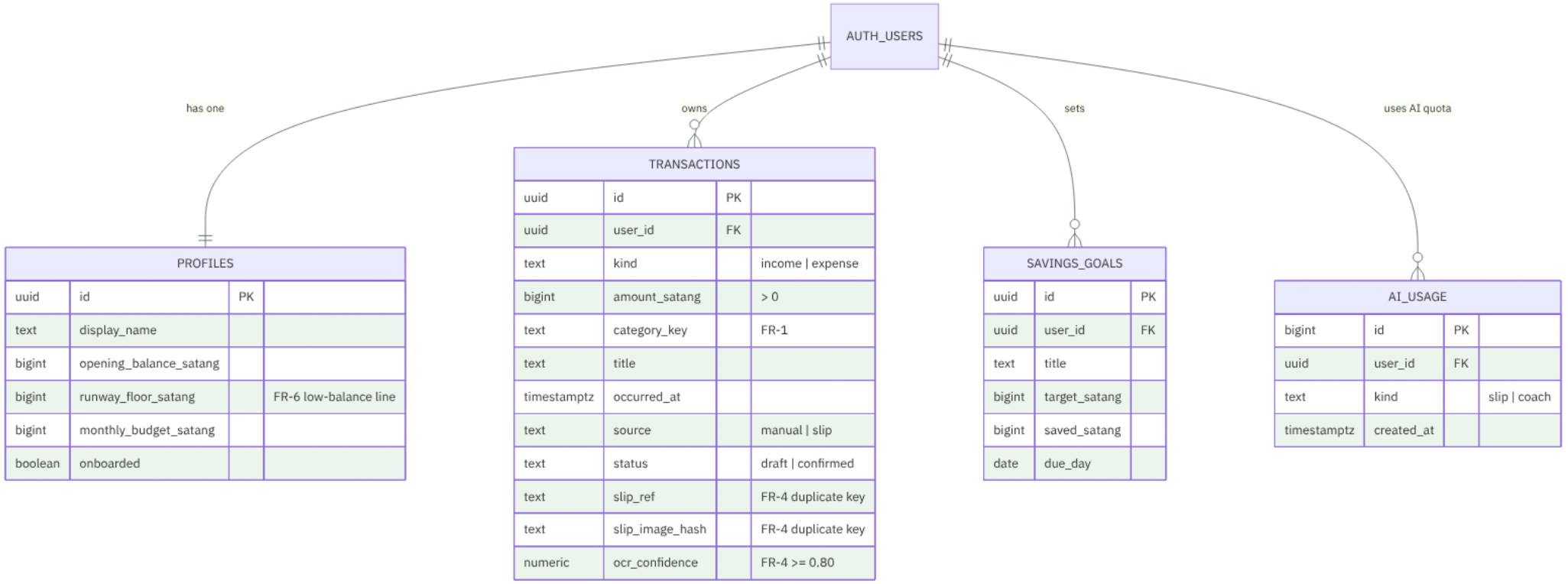
Supabase: email sign-in, PostgreSQL with row-level security

Edge Functions call the Google Gemini API; the key stays on the server

Money stored as whole satang

DESIGN

A diagram for every requirement



ER diagram from our database migrations · plus use case, architecture, screen flow, and one sequence or flow diagram per FR in `docs/design/`

GOLDEN THREAD

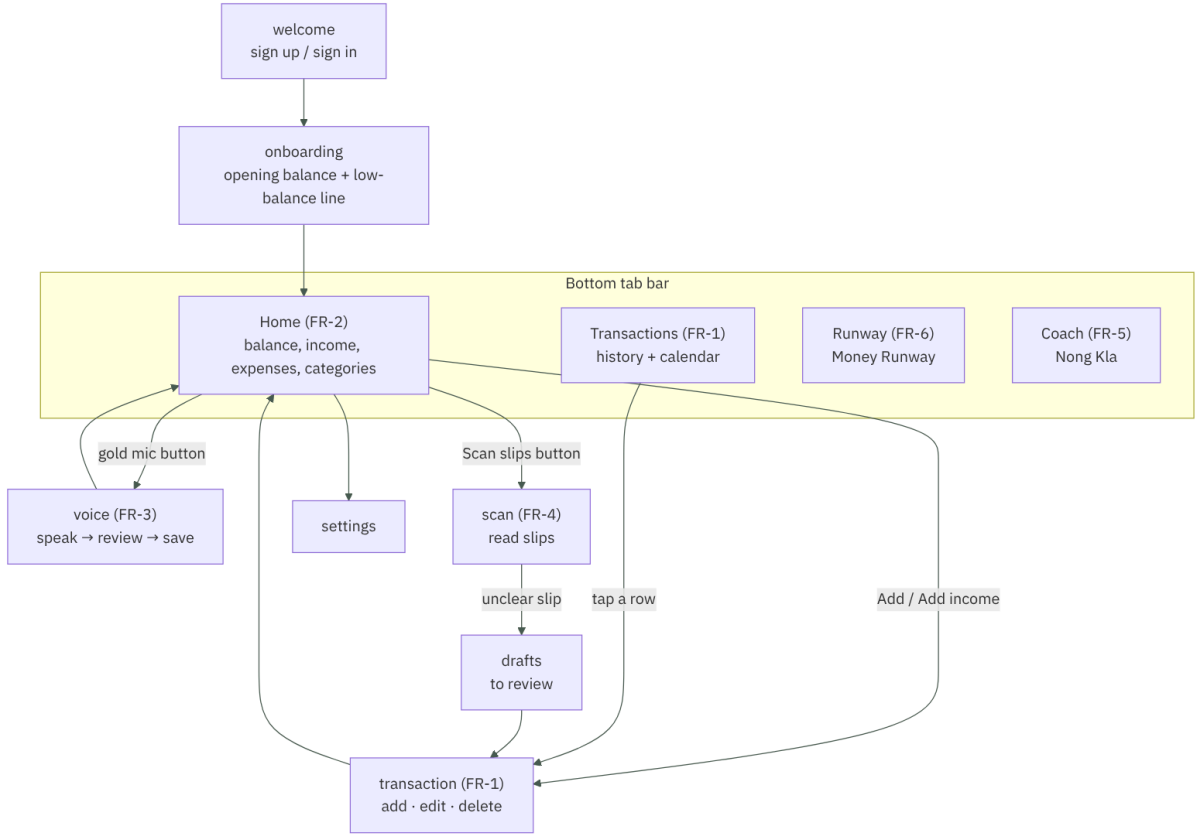
Problem → FR → Design → Prototype → Test

Problem	FR	Design	Prototype	Test
Records incomplete or wrong	**FR-1**	ER, seq 05	Transactions	TC-01, 02, 25, 72, 79 · E2E
Overall status hard to see	**FR-2**	seq 05, flow 04	Home	TC-03, 08–11, 81 · E2E
Manual entry is slow	**FR-3**	seq 06	Voice	TC-59–61, 80 · E2E
Searching and retyping slips	**FR-4**	seq 07, ER	Scan, Drafts	TC-19–30, 64, 83–88 · E2E
Charts don't say what to do	**FR-5**	seq 08	Coach	TC-31, 32, 49–51, 89 · E2E
Balance hides spending pace	**FR-6**	flow 09	Runway	TC-12–18 · E2E

Full table with file paths and manual tests: `docs/final-report.pdf`

PROTOTYPE

Where each requirement lives in the app



FR-1	Transactions tab, Add screen
FR-2	Home tab
FR-3	Mic button on Home
FR-4	Scan slips button, Drafts
FR-5	Coach tab
FR-6	Runway tab, card on Home

FR-6 MONEY RUNWAY

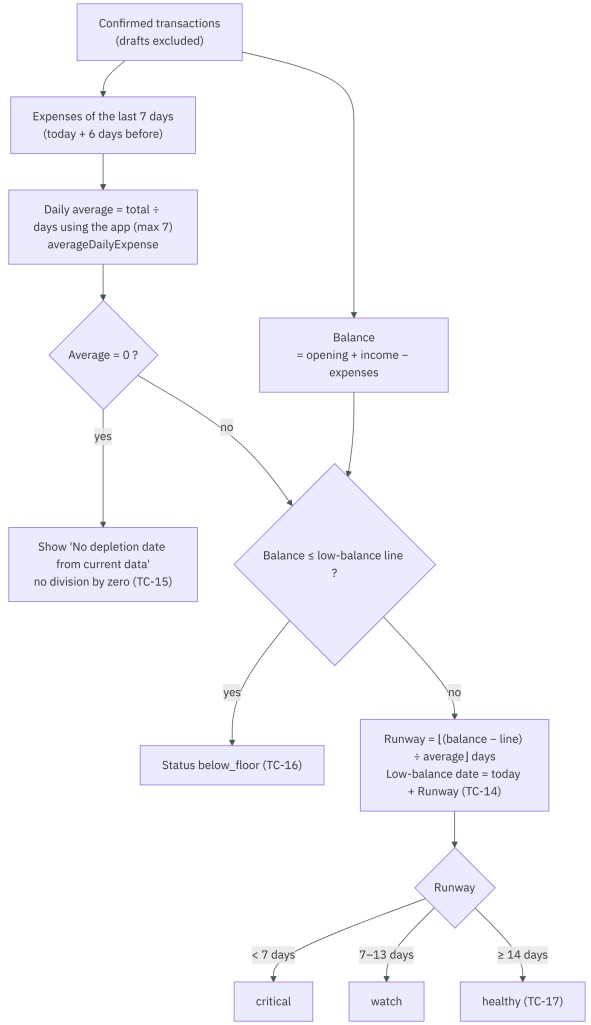
How the low-balance date is calculated

Daily average = confirmed expenses of the last 7 days ÷ days using the app

Runway = (balance – low line) ÷ daily average

If the average is zero, we show "No depletion date from current data" instead of dividing by zero.

Unit tests TC-12 to TC-18



LIVE DEMO

What we will click through

1	Sign in, check Home: balance, income, expenses, categories	FR-2
2	Add ฿85 Food, edit it, delete it, then Undo	FR-1
3	Tap the mic and say "rice 50 baht, water 15"	FR-3
4	Scan a slip: amount, date and bank are read	FR-4
5	Ask the coach "Can I spend 500 baht today?"	FR-5
6	Open Runway and try cutting spending	FR-6

WORTH KNOWING

Security and changes

Security

Each user sees only their own data. The AI key never leaves the server. The coach gets totals only.

Changed by users

One coach persona instead of three tones. Clear slips are counted automatically.

Known limit

Some slips take up to about 50 seconds to read. AI use is logged in `docs/AI_USAGE_LOG.md`.

THANK YOU

Questions?

Web	2550expo-tech.github.io/Socrates-and-Skeletons-
Android	Releases → latest-apk on GitHub
Code	github.com/2550expo-tech/MIndPay